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Widder

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(54) **CORDLESS DRILL POWERED LOG SPLITTER**

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(52) **U.S. Cl.**
CPC **B27L 7/06** (2013.01)

(58) **Field of Classification Search**
CPC B27L 7/00; B27L 7/06; B27L 7/08
See application file for complete search history.

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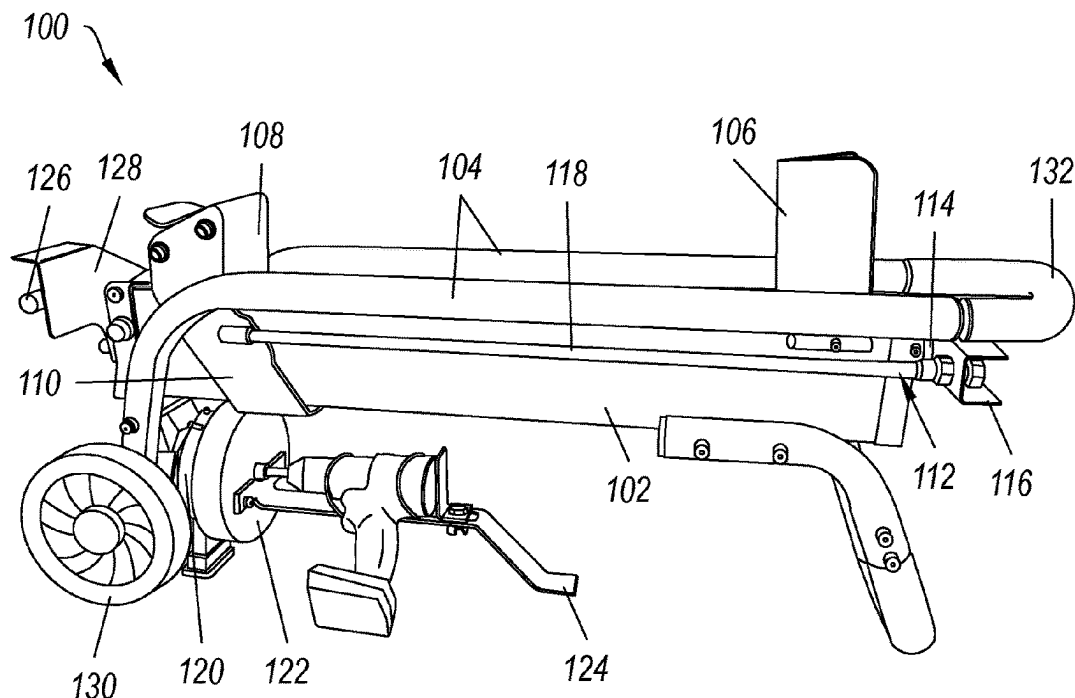
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(57) **ABSTRACT**

A cordless drill powered log splitter. The cordless drill powered log splitter includes a body and one or more rails, the one or more rails configured to support a piece of wood during a splitting operation. The cordless drill powered log splitter also includes a splitting wedge mounted on the body and a contact member, the contact member configured to press the piece of wood against the splitting wedge during the splitting operation. The cordless drill powered log splitter further includes a transmission, the transmission configured to move the contact member relative to the body and connect to a cordless drill, where the cordless drill provides the power to the transmission.

20 Claims, 12 Drawing Sheets



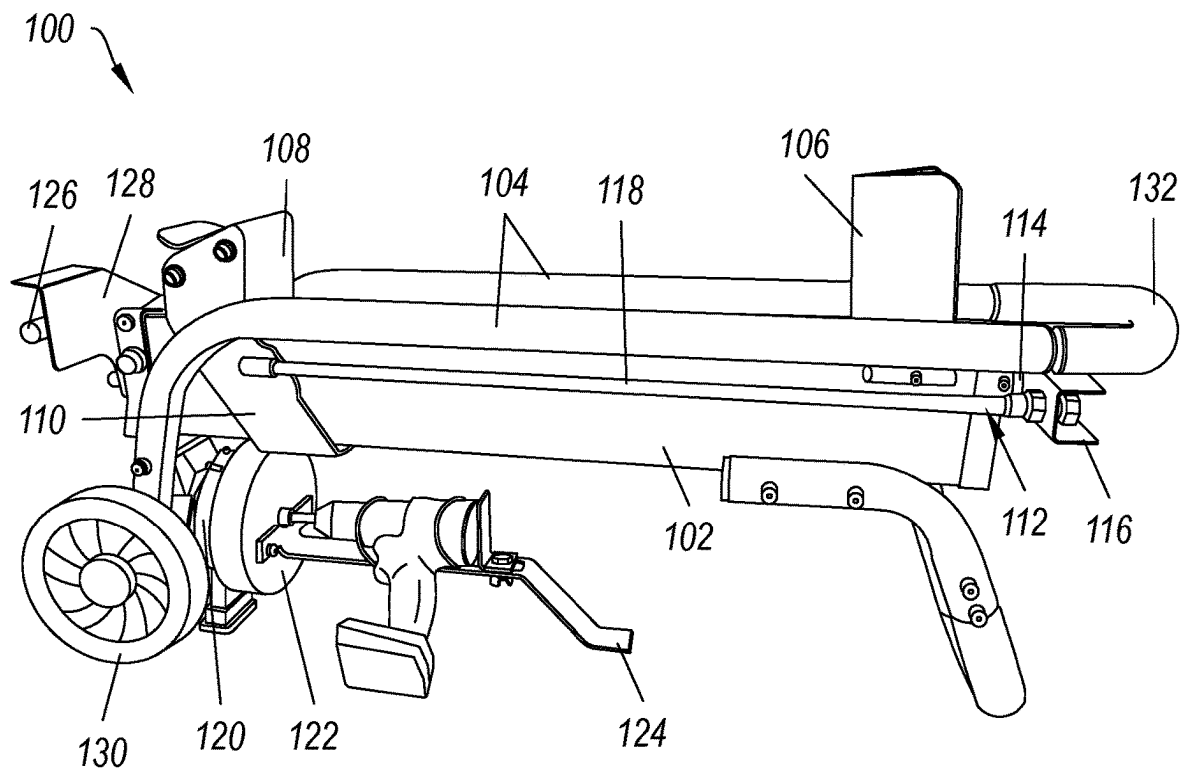


FIG. 1

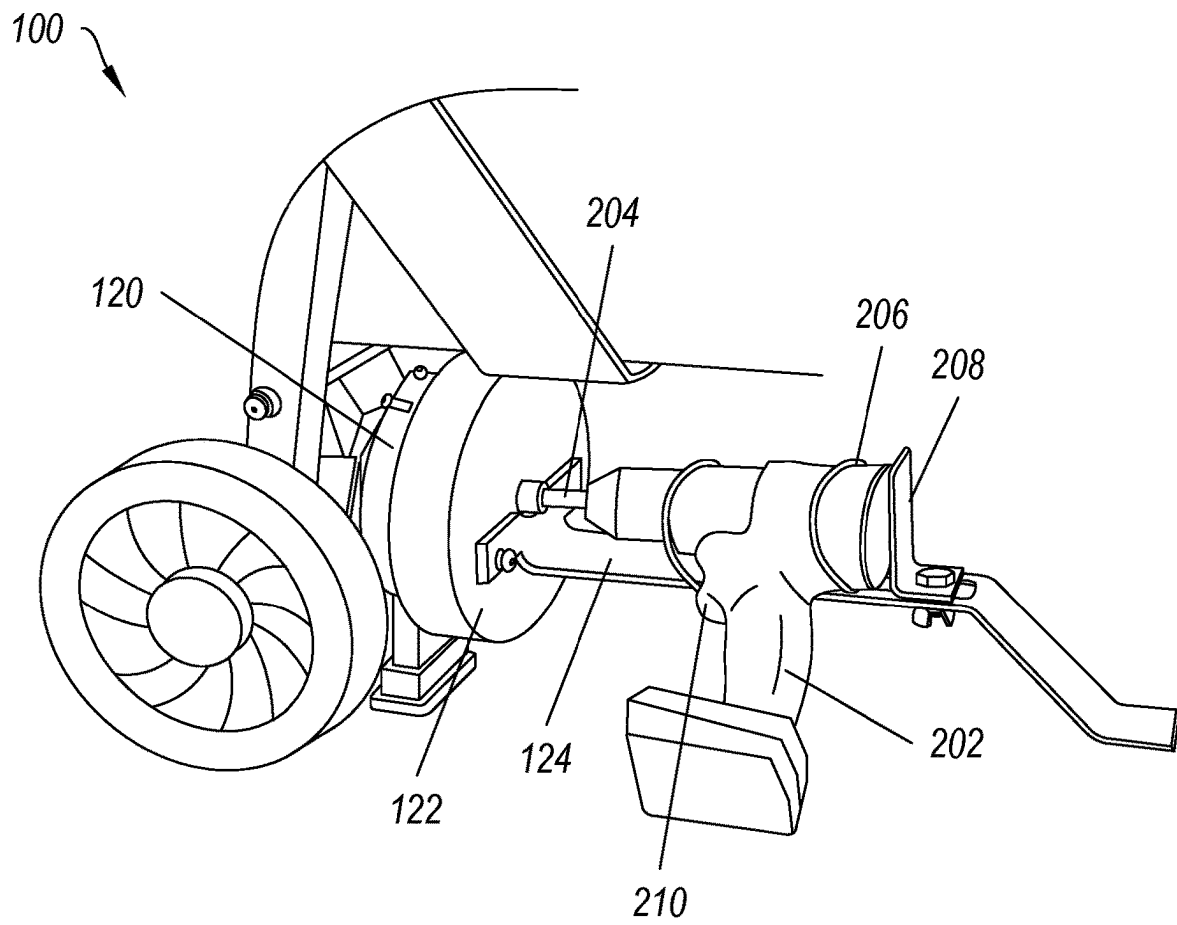


FIG. 2

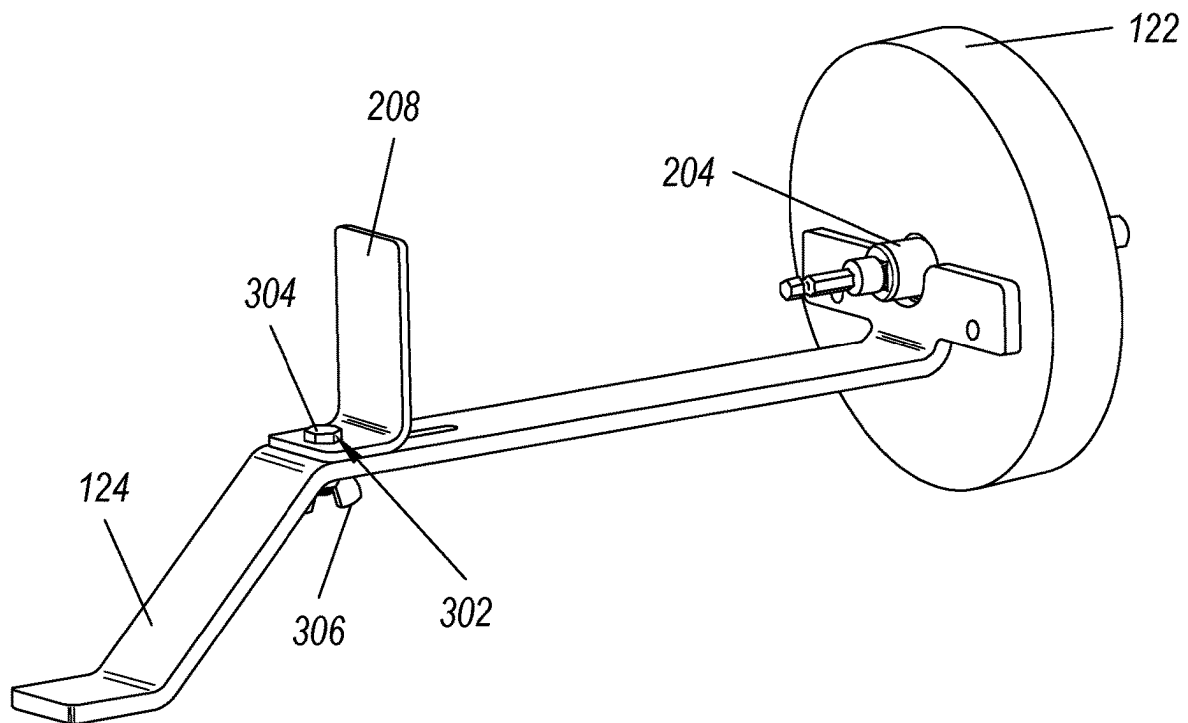


FIG. 3

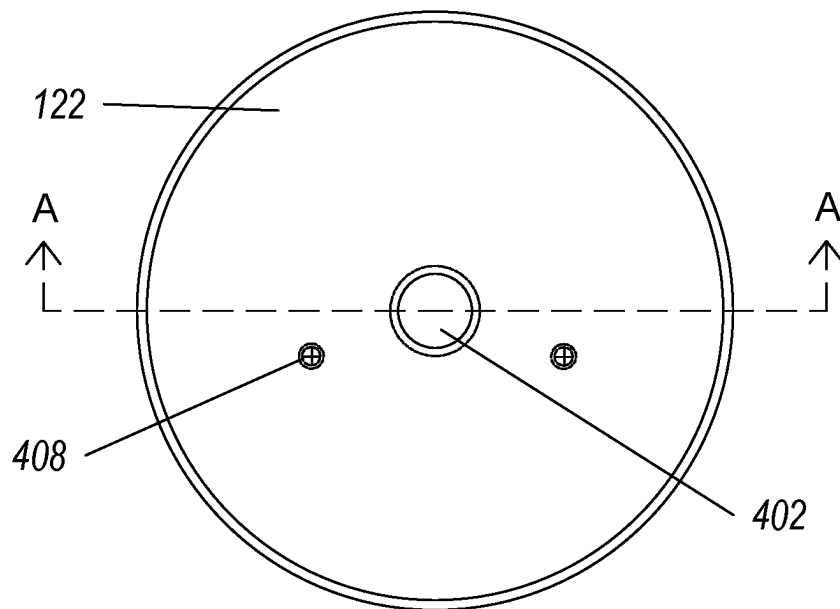


FIG. 4A

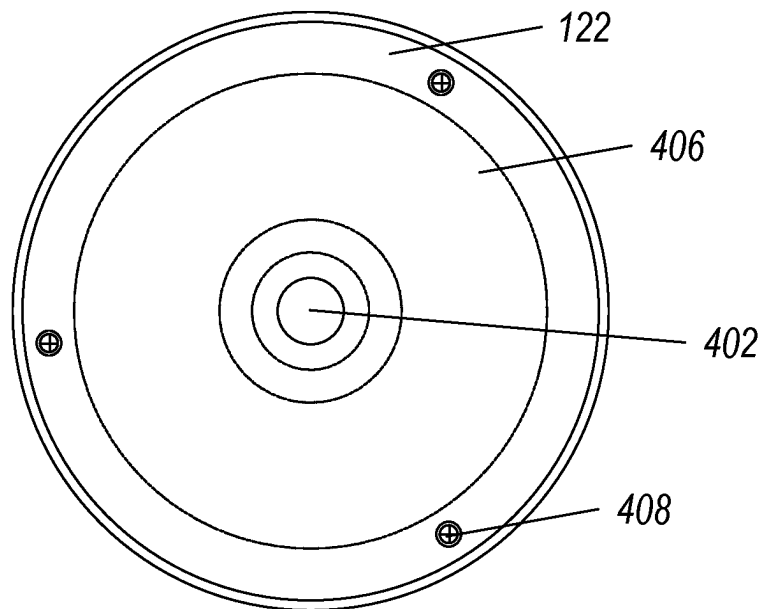


FIG. 4B

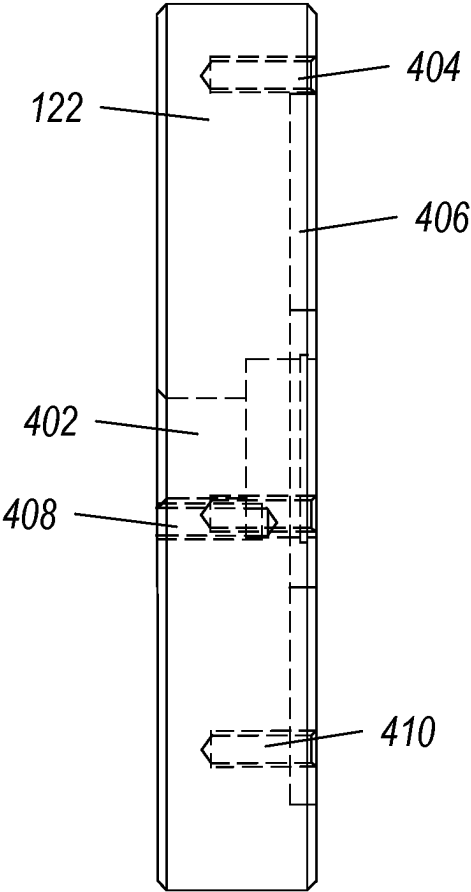


FIG. 4C

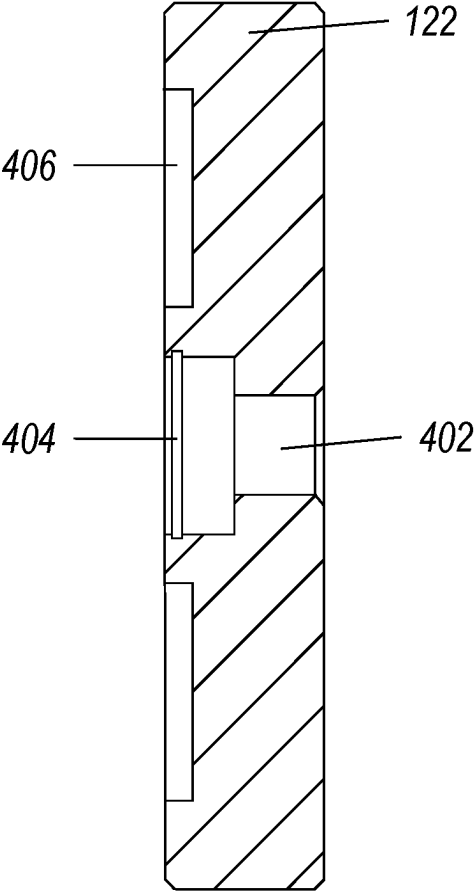


FIG. 4D

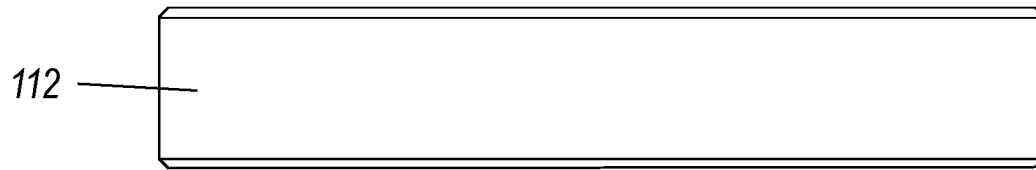


FIG. 4E

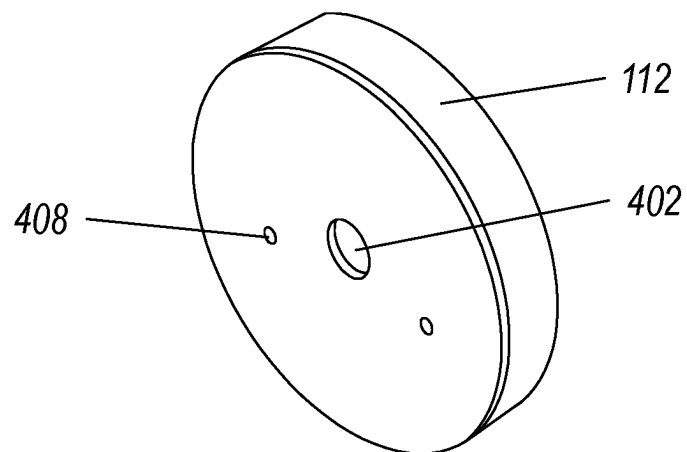


FIG. 4F

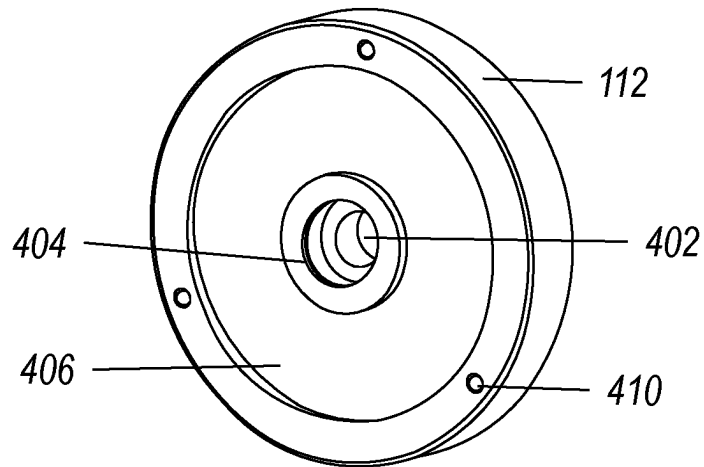


FIG. 4G

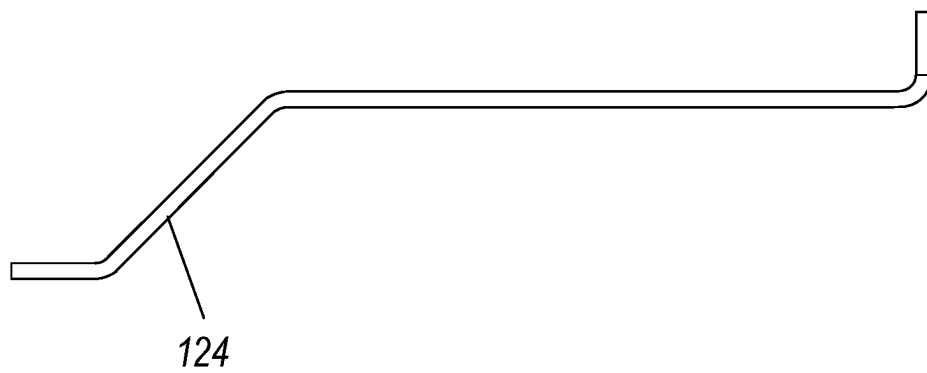


FIG. 5A

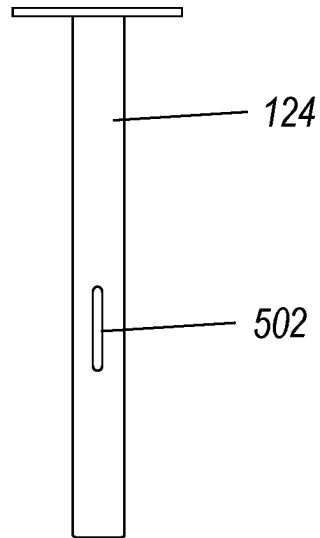


FIG. 5B

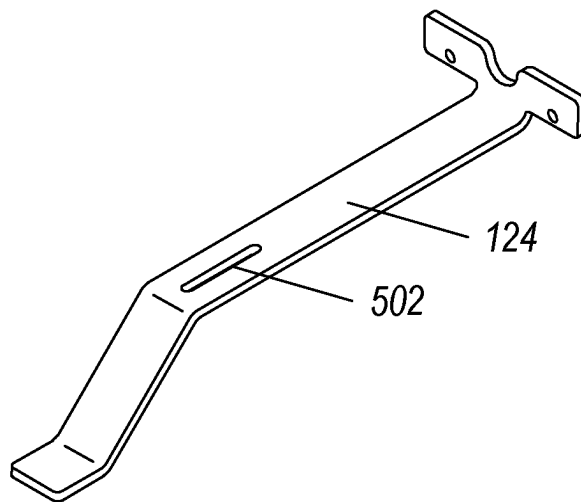


FIG. 5C

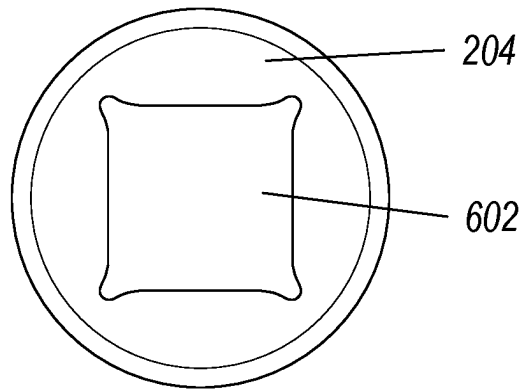


FIG. 6A

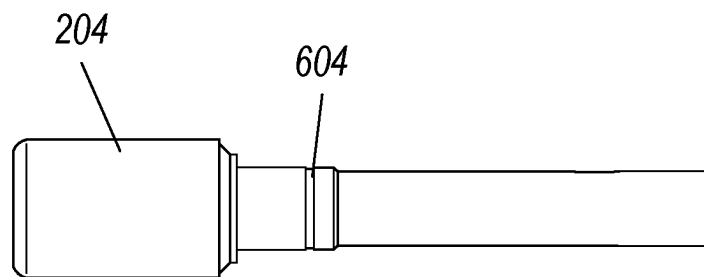


FIG. 6B

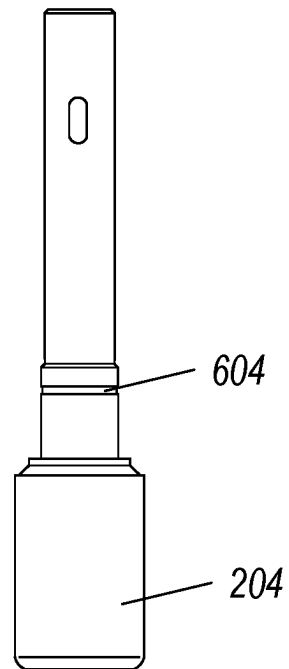


FIG. 6C

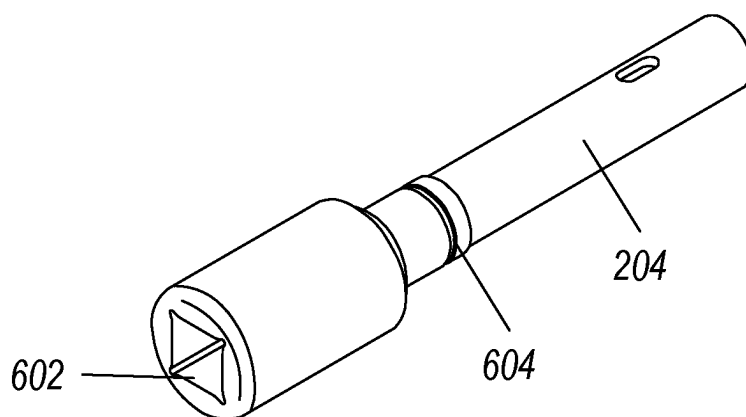


FIG. 6D

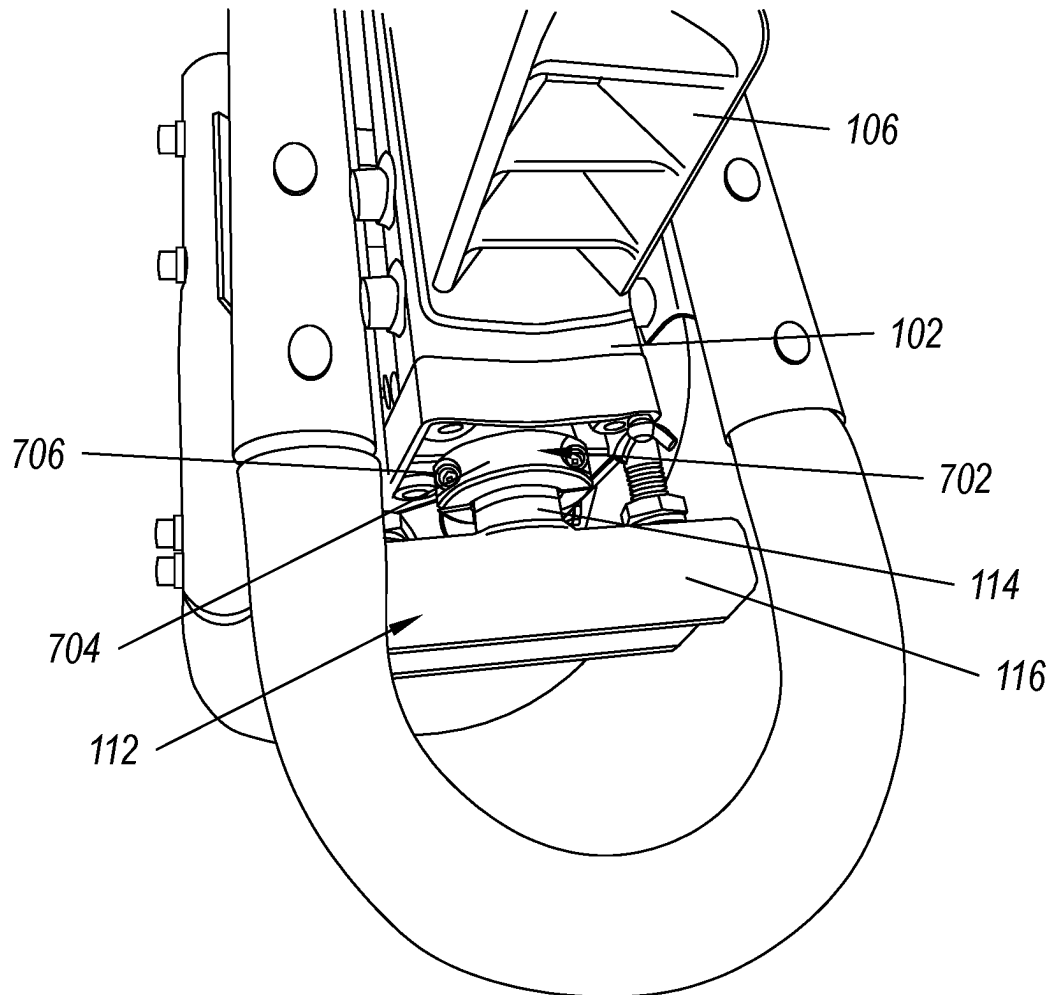


FIG. 7

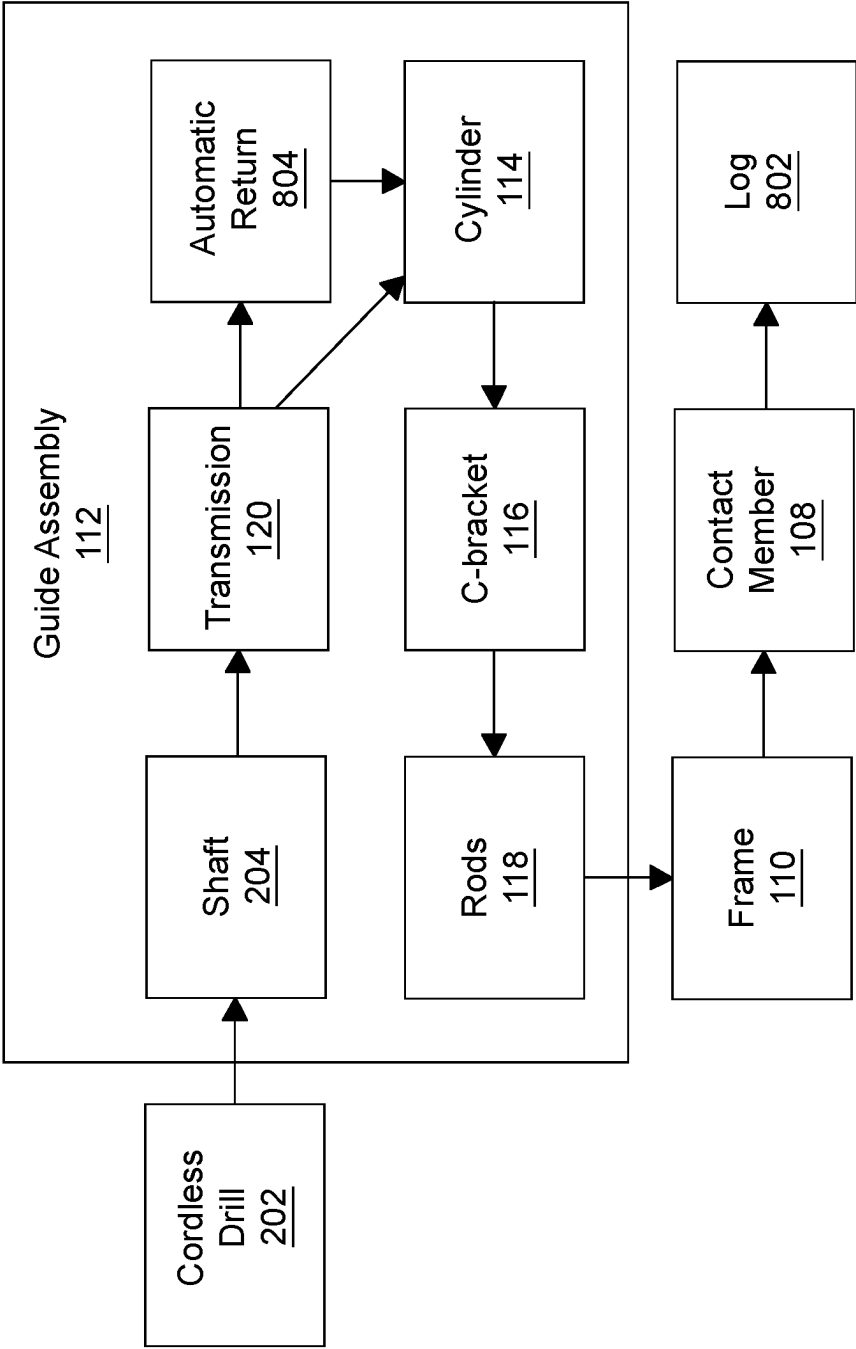


FIG.8

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**CORDLESS DRILL POWERED LOG
SPLITTER****CROSS-REFERENCE TO RELATED
APPLICATIONS**

Not applicable.

BACKGROUND OF THE INVENTION

A log splitter is a piece of machinery or equipment used for splitting firewood from softwood or hardwood logs. There are also manual log splitters, which use mechanical leverage to force logs through a sharpened blade assembly; and screw or 'corkscrew' types that are driven directly from an agricultural tractor's power take-off shaft where the splitter is mounted on the three-point hitch.

Log splitter are generally powered in one of two ways, they are either connected to a power source via an electric cord which powers an electric motor driving a hydraulic pump or by gasoline or diesel engine with or without a tractor. The non-electric versions are intended to be used remotely where the splitter can be moved to the location of the cut wood source. The problem with gas or diesel powered log splitters, however, is that they weigh more than electric log splitters. This is partly due to their more complex engines and partly due to the fact that they have a liquid fuel which must also be transported. Thus, an operator in a remote location must choose between weight and available power sources.

To handle this tradeoff, some operators use an electric log splitter which is powered by a generator or which is plugged into a car or other power source. Overall the weight is greater, but the actual log splitter itself is a lighter version. While this is less efficient, it means that a lighter log splitter can be used in more remote locations.

Therefore, there is a need in the market for a log splitter which is as light as possible, to increase portability but has power sources which are portable as well.

SUMMARY OF THE INVENTION

This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential characteristics of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter.

One example embodiment includes a cordless drill powered log splitter. The cordless drill powered log splitter includes a body and one or more rails, the one or more rails configured to support a piece of wood during a splitting operation. The cordless drill powered log splitter also includes a splitting wedge mounted on the body and a contact member, the contact member configured to press the piece of wood against the splitting wedge during the splitting operation. The cordless drill powered log splitter further includes a transmission, the transmission configured to move the contact member relative to the body and connect to a cordless drill, where the cordless drill provides the power to the transmission.

Another example embodiment includes a cordless drill powered log splitter. The cordless drill powered log splitter includes a body and one or more rails, the one or more rails configured to support a piece of wood during a splitting operation. The cordless drill powered log splitter also

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includes a splitting wedge mounted on the body and a contact member, the contact member configured to press the piece of wood against the splitting wedge during the splitting operation. The cordless drill powered log splitter further includes a transmission, the transmission configured to move the contact member relative to the body and connect to a cordless drill, where the cordless drill provides the power to the transmission. The cordless drill powered log splitter further includes a faceplate, where the faceplate seals the transmission and a shaft, where the shaft passes through the faceplate and is configured to connect to a cordless drill. The cordless drill powered log splitter additionally includes a foot, where the foot is configured to support the cordless drill during operation.

Another example embodiment includes a cordless drill powered log splitter. The cordless drill powered log splitter includes a body and one or more rails, the one or more rails configured to support a piece of wood during a splitting operation. The cordless drill powered log splitter also includes a splitting wedge mounted on the body and a contact member, the contact member configured to press the piece of wood against the splitting wedge during the splitting operation. The cordless drill powered log splitter further includes a frame, where the frame is configured to connect the contact member to the body and a guide assembly. The guide assembly includes a cylinder, the cylinder disposed at least partially within the body and a c-bracket, the c-bracket connect to the cylinder. The guide assembly additionally includes one or more rods, where the one or more rods are connected to the c-bracket at a first end and connected to the frame at a second end. The cordless drill powered log splitter moreover includes a transmission, the transmission configured to move the cylinder relative to the body and a faceplate, where the faceplate seals the transmission. The cordless drill powered log splitter also includes a shaft, where the shaft passes through the faceplate and is configured to connect to a cordless drill. The cordless drill powered log splitter further includes a foot, where the foot is configured to support the cordless drill during operation a switch, where the switch must be actuated in order for the contact member to move relative to the body. The cordless drill powered log splitter additionally includes a stop, where the stop allows the user to set the initial distance between the contact member and the splitting wedge and a return, where the return automatically returns the contact member to its original position after the splitting operation.

These and other objects and features of the present invention will become more fully apparent from the following description and appended claims, or may be learned by the practice of the invention as set forth hereinafter.

BRIEF DESCRIPTION OF DRAWINGS

To further clarify various aspects of some example embodiments of the present invention, a more particular description of the invention will be rendered by reference to specific embodiments thereof which are illustrated in the appended drawings. It is appreciated that these drawings depict only illustrated embodiments of the invention and are therefore not to be considered limiting of its scope. The invention will be described and explained with additional specificity and detail through the use of the accompanying drawings in which:

FIG. 1 illustrates a perspective view of the cordless drill powered log splitter;

FIG. 2 illustrates a closeup view of a cordless drill connected to a cordless drill powered log splitter;

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FIG. 3 illustrates an example of a foot and shaft connected to the faceplate;

FIG. 4A illustrates a front view of the example of a faceplate;

FIG. 4B illustrates a rear view of the example of a faceplate;

FIG. 4C illustrates a side view of the example of a faceplate with internal portions shown in dashed lines;

FIG. 4D illustrates a cut-away view of the example of a faceplate along line A-A of FIG. 4A;

FIG. 4E illustrates a side view of the example of a faceplate;

FIG. 4F illustrates a front left top isometric view of the example of a faceplate;

FIG. 4G illustrates a rear right top isometric view of the example of a faceplate;

FIG. 5A illustrates a side view of the example of a foot;

FIG. 5B illustrates a top view of the example of a foot;

FIG. 5C illustrates an isometric view of the example of a foot;

FIG. 6A illustrates a front view of the example of a shaft;

FIG. 6B illustrates a left-side view of the example of a shaft;

FIG. 6C illustrates a top view of the example of a shaft;

FIG. 6D illustrates an isometric view of the example of a shaft;

FIG. 7 illustrates an example of a stop; and

FIG. 8 is a block diagram that illustrates an example of power being transmitted from a cordless drill to a log through a guide assembly.

DETAILED DESCRIPTION OF THE INVENTION

Reference will now be made to the figures wherein like structures will be provided with like reference designations. It is understood that the figures are diagrammatic and schematic representations of some embodiments of the invention, and are not limiting of the present invention, nor are they necessarily drawn to scale.

FIG. 1 illustrates a perspective view of the cordless drill powered log splitter 100. The cordless drill powered log splitter 100 is designed to replace gas-powered and electrically powered log splitters which are harmful for the environment and more complicated for use. Further, the cordless drill powered log splitter 100 obviates the necessity of having to deal with electrical cords or lower power performance during operation of splitting a log.

Experimentation has shown that the cordless drill powered log splitter 100 has a number of advantages over existing log splitters. For example, the cordless drill powered log splitter 100 can produce $\frac{3}{4}$ of a cord of wood in a single hour, which is comparable to other log splitters. The cord is a unit of measure of dry volume used to measure firewood and pulpwood in the United States and Canada. A cord is the amount of wood that, when "racked and well stowed" (arranged so pieces are aligned, parallel, touching and compact), occupies a volume of 128 cubic feet (3.62 m³). This corresponds to a well-stacked woodpile 4 feet (122 cm) high, 8 feet (244 cm) wide, and 4 feet (122 cm) deep; or any other arrangement of linear measurements that yields the same volume.

However, the weight of the cordless drill powered log splitter 100 is significantly lower than existing log splitters. For example, the cordless drill powered log splitter 100 is approximately 70 pounds, versus 88 pounds for an electric log splitter and 235 pounds for a gas-powered electric log

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splitter. This reduction in weight is due to a number of factors, which are described below. The lower weight increases portability and maneuverability.

FIG. 1 shows that the cordless drill powered log splitter 100 can include a body 102. The body 102 provides durability to the cordless drill powered log splitter 100 for handling large and bulky logs and firewood. Likewise, the body 102 holds the other portions of the cordless drill powered log splitter 100 to ensure the proper configuration of all components. The body 102 can be created from any desired material, such as aluminum, steel, cast iron or polymers. The body 102 can be a hollow rectangular cylinder, allowing other portions of the cordless drill powered log splitter 100 to be placed therein, as described below.

FIG. 1 also shows that the body 102 can include rails 104. The rails 104 run from the front of the body 102 to the rear of the body 102. Further, the rails 104 are configured to support the log during a splitting operation and to provide a guide for the log to move during a splitting operation. The body 102 can be a third support for the wood to be split or the rails 104 can elevate the wood away from the body 102. I.e., the rails 104 can provide a "track" for the wood during splitting which is elevated from the body 102. Because of this, the rails 104 should be strong yet low friction, such as tubular steel (hollow steel tubes), aluminum, or other materials. Tubular steel provides rails 104 that are strong but lighter than solid steel. In addition, coatings can be present which decrease the friction on the rails 104. For example, coatings can be polymers, paint, silicones, lubricants or any other desired coating. As used in the specification and the claims, the phrase "configured to" denotes an actual state of configuration that fundamentally ties recited elements to the physical characteristics of the recited structure. That is, the phrase "configured to" denotes that the element is structurally capable of performing the cited element but need not necessarily be doing so at any given time. Thus, the phrase "configured to" reaches well beyond merely describing functional language or intended use since the phrase actively recites an actual state of configuration.

An additional benefit of the rails 104 is that they form a framework to increase the stability and portability of the cordless drill powered log splitter 100. In particular, the rails 104 can be used by a user to pick up the cordless drill powered log splitter 100. Because the cordless drill powered log splitter 100 is lighter than other log splitters, it can easily be lifted by the rails 104 by a single user. In addition, the rails 104 form a scaffold or skeleton which is rigid enough that it can strengthen the cordless drill powered log splitter 100.

FIG. 1 additionally shows that the cordless drill powered log splitter 100 can include a splitting wedge 106. The splitting wedge 106 is securely attached to the body 102. The splitting wedge 106 is a triangular-shaped member having a vertical edge directed toward the longitudinal direction of the body 102 such that the vertical edge is configured to contact and cut a log. The splitting wedge 106 has a pair of side pieces that extend from the vertical edge and form an angle therebetween. The angle formed between the side pieces can vary. The preferred angle formed by the side pieces is between about thirty and fifty-five degrees (30°-55°). Because the splitting wedge 106 is subject to large forces, it should be made of durable materials such as steel, cast iron, aluminum, or polymers.

FIG. 1 moreover shows that the cordless drill powered log splitter 100 can include a contact member 108. The contact member 108 can include a plate that is configured to press on the log. That is, the contact member 108 is configured to

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move, putting force on the log, pressing the log onto the splitting wedge **106**. I.e., the contact member **108** presses the log onto the splitting wedge **106** with enough force that the log is split by the splitting wedge **106**. Thus, the contact member **108** needs to be sturdy enough that the resistant forces of the log don't cause damage to the contact member **108**.

FIG. 1 also shows that the contact member **108** can include a frame **110**. The frame **110** is placed around the body **102**, ensuring that the contact member **108** can move longitudinally relative to the body **102**, but does not separate from the body **102**. Wheels or bearings can be placed between the frame **110** and the body **102** to allow the frame **110** to move longitudinally relative to the body **102**. A bearing is a machine element that constrains relative motion to only the desired motion, and reduces friction between moving parts. Therefore, bearings allow the frame **110** to move longitudinally relative to the body **102** while remaining aligned axially with the body **102**.

FIG. 1 also shows that the cordless drill powered log splitter **100** can include a guide assembly **112**. The guide assembly **112** is configured to actuate the contact member **108** toward and away from the stationary splitting wedge **106** in a translational manner by way of reciprocal movement relative to the longitudinal direction of the body **102** and the rails **104**. I.e., the guide assembly **112** moves the contact member **108** relative to the body **102**. That is, the guide assembly **112** provides the force for the contact member **108** to drive the log onto the splitting wedge **106**, splitting the wood.

FIG. 1 further shows that the guide assembly **112** can include a cylinder **114**. The cylinder **114** is disposed within the body **102** and is configured to extend and retract actuating the contact member **108** toward and away from the splitting wedge **106**. The movement of the cylinder **114** can be by any desired actuating means. For example, the cylinder **114** can be moved by a hydraulic system, compressed air, a screw drive or any other desired drive system.

FIG. 1 additionally shows that the guide assembly **112** can include a c-bracket **116**. The c-bracket **116** is connected to the cylinder **114**. When the cylinder **114** is actuated such that the cylinder **114** extends from the body **102**, the c-bracket **116** is driven away from the body **102** and when the cylinder **114** is actuated such that cylinder **114** retracts into the body **102**, the c-bracket **116** is pulled toward the body **102**.

FIG. 1 moreover shows that the guide assembly **112** can include rods **118**. The rods **118** are connected to the c-bracket **116** and the contact member **108** or frame **110**. The rods **118** are parallel to the body **102** and the rails **104**. Any movement of the cylinder **114** is transmitted through the c-bracket **116** and the rods **118** to the contact member **108** (directly or through the frame **110**) driving the contact member **108** toward the splitting wedge **106** or away from the splitting wedge **106**. This allows a tremendous amount of force to be applied to the contact member **108** as needed to drive the log onto the splitting wedge **106**.

FIG. 1 also shows that the cordless drill powered log splitter **100** can include a transmission **120** (aka "geared pump" or "transfer"). The transmission **120** is connected to the cylinder **114**, actuating the cylinder **114** relative to the body **102**. I.e., force in the transmission **120** is transmitted to the cylinder **114** which moves the c-bracket **116** and connected rods **118**, frame **110** and the contact member **108**, moving the contact member **108**. The transmission **120** is connected to a cordless drill, as described below.

The transmission can include a drive shaft. A drive shaft is a component for transmitting mechanical power and

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torque and rotation. A drive shaft is used to connect components that cannot be connected directly because of distance or the need to allow for relative movement between them. I.e., the driveshaft is the portion of the transmission that carries the torque from the cordless drill to the cylinder **114**. As torque carriers, drive shafts are subject to torsion and shear stress, equivalent to the difference between the input torque and the load. They must therefore be strong enough to bear the stress, while avoiding too much additional weight as that would in turn increase their inertia. The driveshaft can include one or more universal joints, jaw couplings, or rag joints, and sometimes a splined joint or prismatic joint.

FIG. 1 further shows that the cordless drill powered log splitter **100** can include a faceplate **122**. The faceplate **122** prevents direct access to the transmission **120**. I.e., the faceplate **122** helps seal off the transmission **120** while allowing power to be transmitted from the cordless drill to the transmission **120**, as described below. This means that the transmission **120** cannot lose oil and is protected from dirt and debris buildup. This is critical to reduced maintenance and increased reliability. I.e., since there are not oil changes or other maintenance required portability and reliability is increased.

FIG. 1 additionally shows that the cordless drill powered log splitter **100** can include a foot **124**. The foot **124** is configured to support the cordless drill and make connection between the cordless drill and the faceplate **122** easier. Further, the foot **124** adds stability, allowing the cordless drill powered log splitter **100** to be placed on a raised surface, such as a workbench or truck bed, during operation. Additionally, the foot **124** helps secure the cordless drill in place, as described below.

FIG. 1 additionally shows that the cordless drill powered log splitter **100** can include a switch **126**. The switch **126** is a critical safety feature. In order to operate the cordless drill powered log splitter **100**, the user needs to actuate the switch **126** and pull the trigger on the connected cordless drill. This two-handed operation ensures that the user cannot accidentally turn on the cordless drill powered log splitter **100**, greatly reducing the chances for an accident.

FIG. 1 moreover shows that the cordless drill powered log splitter **100** can include a guard **128**. The guard **128** prevents the user's hands from being near the contact member **108** during operation. I.e., the user's hands are on the cordless drill trigger and the switch **126** and the guard **128** prevents the user's hand on the switch from slipping and coming into contact with any moving parts. The guard **128** also prevents debris from a log which is being split from flying toward the user's hand which is operating the switch **126**.

FIG. 1 also shows that the cordless drill powered log splitter **100** can include a pair of wheels **130**. The wheels **130** allow ease of portability. I.e., the wheels **130** allow the cordless drill powered log splitter **100** to be easily moved to different work areas. The wheels **130** are large enough that the cordless drill powered log splitter **100** can be moved in outdoor areas.

FIG. 1 further shows that the cordless drill powered log splitter **100** can include a handle **132**. The handle **132** is used for moving the drill powered log splitter **100**. I.e., a user can lift the handle **132**, which means the majority of the weight of the cordless drill powered log splitter **100** moves to the wheels **130**. The cordless drill powered log splitter **100** can then be wheeled to the desired position and the handle **132** lowered until the cordless drill powered log splitter **100** rests on the ground. In addition, because the handle **132** is connected to the rails **104**, the combination of the handle and

the rails **104** can be used to lift the entirety of the cordless drill powered log splitter **100**.

FIG. 2 illustrates a closeup view of a cordless drill **202** connected to a cordless drill powered log splitter **100**. The cordless drill **202** is connected via the faceplate **122** and is secured in place on the foot **124**. Cordless drills **202** are ubiquitous and are portable, compact and light. They are easy to operate and can have batteries changed easily. The cordless drill **202** has a motor that is driven by electricity to produce a rotary motion. This means that the cordless drill powered log splitter **100** does not need a dedicated motor.

FIG. 2 shows that the cordless drill **202** is connected to a shaft **204**. The shaft **204** passes through the faceplate **122** into the transmission **120**. Thus, any torque produced by the cordless drill **202** is transmitted to the shaft **204** and from there to the transmission **120**, which ultimately results in force being applied to the contact member which pushes a log onto the splitting wedge. The shaft **204** passes through the faceplate **122** which means that the transmission **120** remains sealed, reducing wear and maintenance needs.

FIG. 2 also shows that cordless drill powered log splitter **100** can include an attachment **206**. The attachment **206** secures the cordless drill **202** in place during operation. Specifically, the attachment **206** can ensure that the body of the cordless drill **202** rests on the foot **122** to reduce wobble and make it easy for an operator to keep the drill aligned with the shaft **204** while the motor of the cordless drill **202** is engaged. The attachment **206** can include any desired material, such as zip ties, cords, bungee cords or any other desired attachment mechanism. The attachment **206** can wrap around the foot **122** or can be otherwise secured to the foot **122**.

FIG. 2 further shows that the cordless drill powered log splitter **100** can include a bracket **208**. The bracket **208** keeps the cordless drill **202** engaged with the shaft **204**. I.e., the cordless drill **202** cannot be pulled out of the shaft **204** because of the bracket **208**. Thus, the attachment **206** and the bracket **208** work together to ensure that the user does not have to align the drill with the shaft **204** during operation. This allows the user to pull the trigger **210** of the cordless drill while holding the switch **126**, increasing user safety.

FIG. 3 illustrates an example of a foot **124** and shaft **204** connected to the faceplate **122**. The foot **124**, shaft **204** and faceplate **122** are critical to allow for the cordless drill powered log splitter to be as portable as possible. This is for a number of reasons. First, the foot **124**, shaft **204** and faceplate **122** dramatically reduce the weight of the cordless drill powered log splitter. In particular, the foot **124**, shaft **204** and faceplate **122** eliminate the need for a motor, which is a large portion of the weight. It also eliminates the need for extension cords or other accessories.

Additionally, the foot **124**, shaft **204** and faceplate **122** reduce the number of components under the body, which allows the cordless drill powered log splitter to be maneuvered in a number of different ways. For example, a bracket can be placed under the body **102**, which is made possible by the absence of a motor and mounted to a hitch or other mechanical means for transporting the cordless drill powered log splitter. I.e., the cordless drill powered log splitter can be mounted to sit parallel to the tailgate of a truck for transport and can be used in place (i.e., it does not have to be removed from the bracket for use). Similar mounting can be done on other vehicles, such as an ATV, camper, etc. Likewise, the absence of a motor creates a "cavity" under the body **102** which allows the cordless drill powered log splitter to be placed on a user's shoulder to be walked to an area for use. This means that the cordless drill powered log

splitter can be transported with ease or can be used in places such as workbenches with ease, giving a user maximum flexibility and portability.

FIG. 3 shows that the bracket **208** can be secured using a fastener **302**. The fastener **302** can be loosened to allow the bracket **208** to move laterally along the foot **122**. The fastener **302** can likewise be tightened when the bracket **208** is pushed against the back of the cordless drill to keep the cordless drill engaged with the shaft **204**. I.e., loosening the fastener **302** and the attachment allows the cordless drill **202** to be disengaged from the cordless drill powered log splitter **100**.

Exemplarily, FIG. 3 shows a bolt **304** and wingnut **306** as the fastener. It is critical that the fastener **302** be securable without tools. Toolless operation increases portability because it allows the cordless drill to be secured to the cordless drill powered log splitter **100** on site, which reduces the weight of the cordless drill powered log splitter. That is, the cordless drill is the motor for the cordless drill powered log splitter and the power source, which reduces weight and dramatically increases portability. This portability is hampered by requiring tools to secure the fastener **302**.

FIGS. 4A-4G (collectively "FIG. 4") illustrates an example of a faceplate **122**. FIG. 4A illustrates a front view of the example of a faceplate **122**; FIG. 4B illustrates a rear view of the example of a faceplate **122**; FIG. 4C illustrates a side view of the example of a faceplate **122** with internal portions shown in dashed lines; FIG. 4D illustrates a cut-away view of the example of a faceplate **122** along line A-A of FIG. 4A; FIG. 4E illustrates a side view of the example of a faceplate **122**; FIG. 4F illustrates a front left top isometric view of the example of a faceplate **122**; and FIG. 4G illustrates a rear right top isometric view of the example of a faceplate **122**. The faceplate **122** is critical to allow power transmission to the transmission of the cordless drill powered log splitter while sealing off the transmission. This reduces maintenance and prolongs usability.

FIG. 4 shows that the faceplate **122** can include an aperture **402**. The aperture **402** allows the shaft to pass through the faceplate **122**. The shaft is sealed within the aperture **402**. This prevents leakage of hydraulic fluid from the transmission. I.e., the shaft within the aperture **402** and the faceplate together form a leak proof surface, through which hydraulic fluid cannot pass so that all hydraulic fluid is retained within the transmission. However, the shaft can turn within the aperture **402**. This allows the torque provided a portable drill to pass through the faceplate to the transmission.

The seal between the shaft and the aperture **402** can include any desired sealing mechanism. For example, the seal can be created by an O-ring. The O-ring is placed around the shaft within the aperture **402** in groove **404** so that it can turn within the aperture **402**. I.e., the shaft is not in direct contact with the walls of the aperture **402**, instead it is in contact with an O-ring which is in contact with the groove **404** in the aperture **402**. An O-ring, also known as a packing or a toric joint, is a mechanical gasket in the shape of a torus; it is a loop of elastomer with a round cross-section, designed to be seated in a groove and compressed during assembly between two or more parts, forming a seal at the interface.

Likewise, there is a seal between the faceplate **122** and an attached transmission. This can include a rubber seal, such as an O-ring, or any other desired seal placed in groove **406**. This prevents hydraulic fluid from leaving the transmission.

I.e., it allows the faceplate **122** to seal the hydraulic fluid within the transmission, lowering maintenance costs and increasing reliability.

FIG. **4** also shows that the faceplate **122** can include one or more screw holes **408**. The screw holes **408** allow the foot to be attached to the faceplate **122** in the correct alignment. I.e., the screw holes **408** are positioned such that the foot will be in the optimal place for attachment of the cordless drill to the shaft when the cordless drill is placed on the foot.

The screw holes **408** create an alignment between the cordless drill and the shaft, such that the cordless drill can be attached in line with the shaft. I.e., the screw holes **408** ensure that the body of the foot is approximately 1¼ inches below the center of the aperture **402**. This drop ensures that cordless drills can be attached to the foot in an alignment where the shaft is easily inserted into the chuck of the cordless drill. As used in the specification and the claims, the term approximately shall mean that the value is within 10% of the stated value, unless otherwise specified.

FIG. **4** further shows that the faceplate **122** can include one or more screw holes **410**. The one or more screw holes **410** are used to secure the faceplate **122** to the transmission. I.e., a bolt can pass through the housing of the transmission into the screw holes **410**. As the bolts are tightened they pull the faceplate securely against the transmission creating a seal between the faceplate **122** and the transmission (because of the rubber seal located in the groove **406** discussed above).

The faceplate **112** is a balance between keeping weight low (to increase portability) and being strong (to resist the pressure created in the hydraulic fluid and support the weight of the foot and drill). In order to achieve this balance, the faceplate is made of a lightweight but strong material such as steel or high strength aluminum.

FIGS. **5A-5C** (collectively "FIG. **5**") illustrate an example of a foot **124**. FIG. **5A** illustrates a side view of the example of a foot **124**; FIG. **5B** illustrates a top view of the example of a foot **124**; and FIG. **5C** illustrates an isometric view of the example of a foot **124**. The foot **124** supports the cordless drill during operation and ensures that the cordless drill remains in place relative to the shaft and faceplate.

Experimentation showed that the placement and design of the foot **124** were critical to proper operation. Without a foot **124** alignment of the cordless drill is much more difficult. This is both because the cordless drill moves freely relative to the shaft and because the user had to pick up and place the cordless drill each time which was not efficient and took more time to align.

In addition, after the foot **124** was originally designed, problems were still observed. In particular, with the original foot **124** and attachment, over time the straps would loosen, and the drill would not be secure interrupting operation, creating a need to add the bracket to keep engagement between the cordless drill and the shaft. Addition of the bracket showed that it needed to be adjustable in order to allow for different cordless drills. This led to the development of the slot **502** and the fastener which created adjustability.

FIGS. **6A-6D** (collectively "FIG. **6**") illustrate an example of a shaft **204**. FIG. **6A** illustrates a front view of the example of a shaft **204**; FIG. **6B** illustrates a left-side view of the example of a shaft **204**; FIG. **6C** illustrates a top view of the example of a shaft **204**; and FIG. **6D** illustrates an isometric view of the example of a shaft **204**. The shaft **204** allows torque to pass through the faceplate into the transmission while retaining hydraulic fluid.

FIG. **6** shows that the shaft **204** includes a square female socket **602**. Experimentation with a hex socket showed that the socket wore over time causing slippage. A square female socket **602** reduces wear and prolongs the life of the shaft **204**.

FIG. **6** also shows that the shaft **204** includes a groove **604**. The groove **604** allows for placement of an O-ring, which can be used to create a seal between the shaft **204** and the faceplate. I.e., the groove **604** ensures the correct placement and retention of an O-ring.

FIG. **7** illustrates an example of a stop **702**. The stop **702** reduces the distance between the contact member and the splitting wedge **106**. A reduced distance between the contact member and the splitting wedge **106** reduces the time needed to split logs and the energy used in any single splitting operation. I.e., reducing the distance between the contact member and the splitting wedge **106** means that the guide assembly **112** has to move less in any splitting operation which means that more log splits can be accomplished on a single battery for the cordless drill. Without the stop **702** the splitting operation is much slower and far less efficient as a portion of the battery life is used to move through empty space. Additionally, the lack of a stop **702** creates a danger to the user as he/she may need to place the log in the middle of a splitting operation (i.e., the user may have to engage the switch and the cordless drill, move the contact member to the correct position, place the log, then resume the splitting operation).

FIG. **7** shows that the stop **702** can include a shaft collar **704** on the cylinder **114**. The shaft collar **704** serves as a mechanical stop and stroke limiter. I.e., the shaft collar **704** can prevent the cylinder **114** from being fully retracted into the body **102**. That is, a user can perform a splitting operation and before the cylinder **114** is retracted, set the shaft collar **704** on the cylinder to shorten the stroke length of the cylinder **114**. Then each stroke covers a shorter distance.

FIG. **7** also shows that the stop **702** can include a fastener **706**. The fastener **706** can include a screw or bolt that tightens the shaft collar **704** around the cylinder **114**. For example the fastener **706** can include a set screw which bites into the cylinder **114** when tightened, holding the shaft collar **704** in place. Alternatively, the fastener **706** can include a single screw with either a hinge or slot opposite the screw which allows the shaft collar **704** to open to release pressure on the shaft collar **704**, allowing it to move freely along the cylinder **114**. The single screw may or may not be fully removable from the shaft collar **704**. Conversely, the fastener **706** can include a pair of screws that hold to independent pieces of the shaft collar **704** to one another. A pair of screws allows the shaft collar **704** to be more easily removed from the cylinder **114**. I.e., a pair of screws allows the shaft collar **704** to be placed without access to the end of the cylinder **114** (which is attached to the c-bracket **116**).

FIG. **8** is a block diagram that illustrates an example of power being transmitted from a cordless drill **202** to a log **802** through a guide assembly **112**. Ultimately, the power provided by the cordless drill **202** has to be used to move a log **802** onto a splitting wedge, or vice versa, for the log to be split. The more efficient the power transfer process, the more logs that can be split before the battery needs to be changed.

FIG. **8** shows that the power from the cordless drill **202** is transmitted to the guide assembly **112**. This is done by a direct connection between the cordless drill **202** and the shaft **204** which passes through the faceplate into the transmission **120**. In turn the transmission moves the cylinder **114**

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relative to the body of the cordless drill powered log splitter and some of the power is passed into an automatic return **804**.

The automatic return **804** will return the cylinder **114** to its "home position" (which is set by the position of the stop as discussed above) without using power from the cordless drill for the return operation. I.e., the automatic return stores some of the power transmitted during the splitting operation and uses that stored power to return the cylinder **114** to its home position. This can be accomplished using any mechanism which can store mechanical power. For example, the automatic return **804** can include a spring which is stretched during the splitting operation. When the cylinder **114** is no longer being extruded, the spring can pull the cylinder **114** back into the body of the cordless drill powered log splitter.

FIG. 8 also shows that the extrusion of the cylinder **114** moves the c-bracket **116**. The c-bracket **116** pulls on the attached rods **118** which are also attached to the frame **110**. The frame **110** is attached to the contact member **108** and moves longitudinally along the body. The contact member **108** then presses the log **802** against a splitting wedge, cause the log **802** to split.

The present invention may be embodied in other specific forms without departing from its spirit or essential characteristics. The described embodiments are to be considered in all respects only as illustrative and not restrictive. The scope of the invention is, therefore, indicated by the appended claims rather than by the foregoing description. All changes which come within the meaning and range of equivalency of the claims are to be embraced within their scope.

What is claimed is:

1. A cordless drill powered log splitter, the cordless drill powered log splitter comprising:

- a body;
- one or more rails, the one or more rails configured to support a piece of wood during a splitting operation;
- a splitting wedge mounted on the body;
- a contact member, the contact member configured to press the piece of wood against the splitting wedge during the splitting operation;
- a transmission, the transmission configured to:
 - move the contact member relative to the body; and
 - connect to a cordless drill, wherein the cordless drill provides the power to the transmission; and
- a foot, wherein the foot is configured to support the cordless drill during operation.

2. The cordless drill powered log splitter of claim 1 further comprising one or more wheels, wherein the one or more wheels:

- are connected to the body; and
- configured to support the weight of the cordless drill powered log splitter.

3. The cordless drill powered log splitter of claim 1 further comprising a guard, the guard configured to prevent the user's hands from being near the splitting wedge during operation.

4. The cordless drill powered log splitter of claim 1, wherein the body is a hollow rectangular cylinder.

5. The cordless drill powered log splitter of claim 4, wherein the one or more rails are parallel to a main axis of the body.

6. The cordless drill powered log splitter of claim 1, wherein the one or more rails includes tubular steel.

7. The cordless drill powered log splitter of claim 1, wherein the contact member includes a plate configured to press on a log.

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8. The cordless drill powered log splitter of claim 1, wherein the one or more rails includes a pair of rails.

9. A cordless drill powered log splitter, the cordless drill powered log splitter comprising:

- a body;
- one or more rails, the one or more rails configured to support a piece of wood during a splitting operation;
- a splitting wedge mounted on the body;
- a contact member, the contact member configured to press the piece of wood against the splitting wedge during the splitting operation;
- a transmission, the transmission configured to:
 - move the contact member relative to the body; and
 - connect to a cordless drill, wherein the cordless drill provides the power to the transmission;
- a faceplate, wherein the faceplate seals the transmission;
- a shaft, wherein the shaft:
 - passes through the faceplate; and
 - is configured to connect to a cordless drill; and
- a foot, wherein the foot is configured to support the cordless drill during operation.

10. The cordless drill powered log splitter of claim 9, wherein the foot includes a bracket, the bracket configured to keep the cordless drill engaged with the shaft.

11. The cordless drill powered log splitter of claim 10 further comprising a fastener, wherein the fastener is configured to prevent movement of the bracket relative to the foot.

12. The cordless drill powered log splitter of claim 11, wherein the fastener is toolless.

13. The cordless drill powered log splitter of claim 12, wherein the fastener includes a bolt and a wingnut.

14. A cordless drill powered log splitter, the cordless drill powered log splitter comprising:

- a body;
- one or more rails, the one or more rails:
 - attached to the body; and
 - configured to support a piece of wood during a splitting operation;
- a splitting wedge mounted on the body;
- a contact member, the contact member configured to press the piece of wood against the splitting wedge during the splitting operation;
- a frame, wherein the frame is configured to connect the contact member to the body;
- a guide assembly, wherein the guide assembly includes:
 - a cylinder, the cylinder disposed at least partially within the body;
 - a c-bracket, the c-bracket connect to the cylinder;
 - one or more rods, wherein the one or more rods are:
 - connected to the c-bracket at a first end; and
 - connected to the frame at a second end;
- a transmission, the transmission configured to move the cylinder relative to the body; and
- a faceplate, wherein the faceplate seals the transmission;
- a shaft, wherein the shaft:
 - passes through the faceplate; and
 - is configured to connect to a cordless drill; and
- a foot, wherein the foot is configured to support the cordless drill during operation; and
- a switch, wherein the switch must be actuated in order for the contact member to move relative to the body;
- a stop, wherein the stop allows the user to set the initial distance between the contact member and the splitting wedge; and

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a return, wherein the return automatically returns the contact member to its original position after the splitting operation.

15. The cordless drill powered log splitter of claim **14**, wherein the return includes a spring. 5

16. The cordless drill powered log splitter of claim **14**, wherein the one or more rods includes a pair of rods.

17. The cordless drill powered log splitter of claim **14**, the frame is placed around the body.

18. The cordless drill powered log splitter of claim **14**, 10 wherein the cylinder is hydraulically driven.

19. The cordless drill powered log splitter of claim **14**, wherein the transmission includes a drive shaft.

20. The cordless drill powered log splitter of claim **14** wherein the one or more rails create a framework which is 15 configured to allow a user to pick up the cordless drill powered log splitter.

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